Sr.No. 8514

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025		Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am		C. Code: CS413
Exam Type: Written	Block:401	Course Section:1	Seat No: 2022600057

1_29_514_986_1_1505_38420

Name of the candidate. Nimit Sheth

Candidate's UID number: 2 0 2 2 6 0 0 0 5 7

Examination class room number: 4 0 1

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)

SEMESTER : 7

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE-AIML

DAY AND DATE: 8/10/25 Wed

COURSE NAME: DL

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

RE/2025/03

8214

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q1) ~~(a) when model~~

Q.1(b) When the model ~~cappa~~ capacity is too high it starts learning the noise in the data and starts memorizing it instead of learning underlying patterns.

→ Since it memorizes the input data, it will perform well on the test set but when new data is introduced it cannot generalize because it has not learnt patterns.

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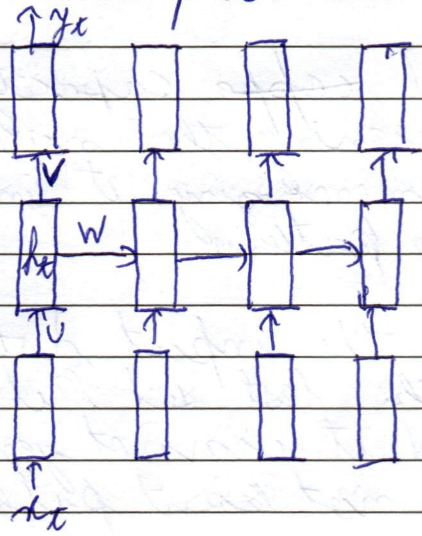
Q2a) ~~The computational complexity~~

Q.3a)



This is basically ~~an R~~ a vanilla RNN at each Timestep

$h_t = Wh_{t-1} + Ux_t + b$

The Loss function is derived from this h_t.

~~$\therefore \frac{\partial L}{\partial W} =$~~

Also since multiple timesteps are involved loss will be the summation of each timestep loss

$\therefore \frac{\partial L}{\partial W} = \sum_{t=1}^{T} \frac{\partial L_t}{\partial W}$

Lets calculate the individual time step gradient In the above example we have four timesteps ~~Let~~ For simplicity we will ~~no~~ calculate gradient for fourth timestep

$\frac{\partial L_4}{\partial W} = \frac{\partial L_4}{\partial h_4} \times \frac{\partial h_4}{\partial W}$

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The term $\frac{\partial h_4}{\partial W}$ will have two parts:

(i) Explicit: Here we consider W to be the only variable and others are constant

(ii) Implicit: Here we consider other variables like h too.

$$\therefore \frac{\partial h_4}{\partial W} = \underbrace{\frac{\partial^+ h_4}{\partial W}}_{\text{Explicit}} + \underbrace{\frac{\partial h_4}{\partial h_3} \times \frac{\partial h_3}{\partial W}}$$

Further splitting the implicit part

$$= \frac{\partial^+ h_4}{\partial W} + \frac{\partial h_4}{\partial h_3}\left(\frac{\partial^+ h_3}{\partial W} + \frac{\partial h_3}{\partial h_2} \times \frac{\partial h_2}{\partial W}\right)$$

$$= \frac{\partial^+ h_4}{\partial W} + \frac{\partial h_4}{\partial h_3}\frac{\partial^+ h_3}{\partial W} + \frac{\partial h_4}{\partial h_3} \times \frac{\partial h_3}{\partial h_2} * \frac{\partial^+ h_2}{\partial W} + \frac{\partial h_4}{\partial h_3} \times \frac{\partial h_3}{\partial h_2} * \frac{\partial h_2}{\partial h_1} \times \frac{\partial^+ h_1}{\partial W}$$

For simplicity we will short circuit the paths

$$\therefore \frac{\partial h_4}{\partial W} = \frac{\partial h_4}{\partial h_4} \times \frac{\partial^+ h_4}{\partial W} + \frac{\partial h_4}{\partial h_3}\frac{\partial^+ h_3}{\partial W} + \frac{\partial h_4}{\partial h_2} \times \frac{\partial^+ h_2}{\partial W} + \frac{\partial h_4}{\partial h_1} \times \frac{\partial^+ h_1}{\partial W}$$

$$\therefore \frac{\partial h_t}{\partial W} = \cancel{\frac{\partial h_t}{\partial h_t}} \sum_{k=1}^{t} \frac{\partial h_t}{\partial h_{t-k}} \frac{\partial^+ h_k}{\partial W}$$

$$\therefore \frac{\partial L_t}{\partial W} = \frac{\partial L_t}{\partial h_t} \sum_{k=1}^{t} \frac{\partial h_t}{\partial h_{t-k}} \frac{\partial^+ h_k}{\partial W}$$

This is the backpropogation through time for an RNN

From the above derivation we observe that as the number of timesteps ~~(i.e number~~ i.e. sequence length increases the gradient multiplications keep increasing and after a point

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the effect of previous hidden states becomes negligible. This is the vanishing gradient problem.

→ LSTM introduce the concept of gates to regulate the flow of information in the network.

→ The forget is basically trained in such a way that it only preserves parts of the hidden state in the long term memory while periodically discarding the ones that have reduced relevance.

→ This reduces the amount of information stored & preventing vanishing gradients

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Q.3(i) GRU introduces two gates as opposed to the three gates of LSTM. They are:

(i) Update gate: Decides what part of the previous and current state to store.

(ii) Reset gate: Decides what part of the previous states to discard.

Let r_t be reset gate, & u_t update gate, U & W are weight matrices, x_t is input

$\therefore r_t = \sigma(W_r h_{t-1} + U_r x_t + b_r)$

~~$\therefore$ current state = tanh(W(x_t * h_{t-1}) + (c_t)~~

Normally we write the current state equation as follows

$c_t = \tanh(W h_{t-1} + U x_t + b)$

But we only want to store selective information in the current state so we add the reset gate

$\therefore c_t = \tanh(W(h_{t-1} * r_t) + U x_t + b)$

Now to regulate the ~~infor~~ state propagated to the next time step i.e. h_t we need the update gate for regulation

$\therefore u_t = \sigma(W_u h_{t-1} + U_u x_t + b_u)$

$\therefore h_t = u_t h_{t-1} + (1 - u_t) c_t$

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This shows that the update rule is a convex combination of the previous and candidate hidden state.

In LSTM we have three gates: input, forget and output gate. Their equations are as follows

$i_t = \sigma(W_i h_{t-1} + U_i x_t + b_i)$

$o_t = \sigma(W_o h_{t-1} + U_o x_t + b_o)$

$f_t = \sigma(W_f h_{t-1} + U_f x_t + b_f)$

→ In LSTMs we can clearly see that there are 3 set of parameters to be trained for each gate.

→ Since GRUs combine the input and forget gate into a single update gate there are only 2 set of parameters to be trained while also giving a good longer term representation of previous states.

→ This also improves the efficiency of the network and makes it faster.

→ In real time applications like ~~chatbots~~ conversational chatbots, speed of the output is important along with accuracy. This is where GRUs can come in handy compared to LSTMs

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Q.2a) ~~Let consider a 2D convolution with input size H × W & K filter of size F × F & stride s~~

~~∴ Output dimension will be as follows~~

$$\cancel{W_o = \frac{W - F + 2P}{s} + 1}$$

In

Q.2a) In depthwise seperable convolution another parameter of depth is there which increase the computation C times compared to 2D convolution

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Rough work

r_t

u_t

c_t

h_t

$4 \times W \times F^2 \times S$

$30\times$